

Dissertation Title:
**Semantic Reconstruction of Software Intent: A Cloud-
Hosted Agentic AI Solution for Study of Requirement
Traceability in Production Grade Codebases**

Course No. : CCZG628T

Course Title: Dissertation

Dissertation Done by:

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Degree Program: M-Tech in Cloud Computing

**Research Area: Artificial Intelligence and Cloud
Computing**

Dissertation carried out at:
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1. Broad Area of Work

Requirement traceability and semantic understanding of software systems have emerged as critical research domains in the current era of large-scale and rapidly evolving software engineering. The broad scope of the proposed dissertation encompasses modern advancements in **Agentic Artificial Intelligence**, **Cloud-based Developer Tooling**, and **Semantic Code Intelligence**, aimed at reconstructing the underlying intent behind implementation decisions in production-grade codebases.

The current research project focuses on the following application areas:

- **Semantic Program Comprehension:** To infer and explain the *why* behind source code structures by analyzing implementation patterns, architectural evolution, and contextual rationale.
- **Requirement Traceability Reconstruction:** To establish meaningful mappings between software requirements, issue tickets, commits, and code changes across enterprise development lifecycles.
- **Knowledge Extraction from DevOps Artifacts:** To retrieve and synthesize information from heterogeneous sources such as JIRA issues, Bitbucket repositories, pull requests, and commit histories.
- **Agentic AI-based Automated Reasoning:** To enable intelligent multi-agent systems capable of drawing contextual conclusions about how and why a particular implementation resolves a requirement or defect.
- **Cloud-hosted AI Developer Assistance:** To deliver real-time intent-aware support within code editors, accelerating onboarding, debugging, and feature development in complex industrial software environments.
- **Machine Learning and Large Language Models:** To detect latent semantic relationships between natural language requirements and code evolution, thereby continuously adapting to new project-specific patterns.

2. Background

Software technologies within modern enterprise organizations are evolving rapidly and have become increasingly complex due to the continuous expansion of large-scale production codebases and cloud-native development practices. In today's industrial work environment, software systems are no longer developed as isolated modules; rather, they exist as highly interconnected ecosystems that evolve through frequent commits, requirement modifications, incident resolutions, and collaborative DevOps-driven workflows. Consequently, developers in such organizations are required not only to understand *what* the code performs, but more critically, *why* a particular implementation was introduced and how it aligns with business or operational requirements.

Within the employing organization's development landscape, engineers heavily rely on distributed artifacts such as IT service management, Code version control histories, pull request discussions, internal documentation etc to interpret system behavior and implementation rationale. However, the intent behind code changes is often fragmented across these heterogeneous platforms, making the process of tracing requirements to actual implementations cognitively demanding and time-consuming. This challenge becomes especially significant during onboarding of new joiners, knowledge transfer sessions, defect triaging, and long-term maintenance of legacy components.

An intelligent, cloud-hosted agentic AI solution is therefore highly relevant in the current organizational context. Such a system can act as a semantic intermediary between developers and the enterprise knowledge ecosystem, enabling automated retrieval and reconstruction of implementation intent from commits, issues, and requirement workflows. The proposed conversational, context-aware AI assistant embedded within code editors can substantially reduce comprehension overhead, accelerate project ramp-up, and provide real-time explanations of how and why specific code modifications resolve operational defects or fulfill business requirements. This directly enhances productivity, maintainability, and traceability in prod grade software development environments.

3. Objectives

The objectives of my dissertation work are as follows:

- **Understand the current scenario with respect to requirement traceability and the challenges of intent loss in production-grade software systems**, where implementation rationale is often fragmented across commits, tickets, and developer discussions.
- **Design an intelligent agentic AI framework that acts as a semantic intermediary between developers and distributed software engineering artifacts**, such as IT service management systems, code repositories, pull requests, CI/CD workflows etc. This adds a lot of positive impact on open source projects.
- **Develop a cloud-hosted AI-assisted solution integrated within code editors to reconstruct the “why” aspect of implementation**, enabling developers to understand how and why specific code changes satisfy requirements or resolve defects.
- **Build an intent-aware conversational software assistant capable of retrieving, synthesizing, and explaining contextual knowledge**, including requirement-to-code mappings, historical evolution of features, and traceability narratives.
- **Support developer productivity through automated guidance for issue reproduction, validation, and testing**, ensuring that implementation correctness can be evaluated against original requirements.
- **Validate the proposed prototype through empirical evaluation and user studies**, focusing on onboarding acceleration, reduction of cognitive load, and improvement in comprehension efficiency for new joiners and open-source contributors.

4. Scope of Work

The scope of this dissertation is to design and develop a cloud-hosted, agentic AI-based semantic assistant integrated within modern code editors, which reconstructs and explains the underlying intent behind software implementations in production-grade codebases. The system aims to establish automated requirement traceability by correlating development artifacts such as IT service management tickets, code repository commits, pull requests, issue discussions etc. Thereby providing developers with contextual understanding of *why* specific code changes were introduced, *how* they resolve requirements or defects, and *when* they should be validated through reproduction and testing workflows.

5. Plan of Work

The dissertation work will be executed in a phased manner, ensuring systematic progression from conceptual groundwork to implementation, evaluation, and final submission. The plan is aligned with the official academic milestones including outline submission, midterm report, and final dissertation

Phases	Start Date-End Date	Work to be done
Dissertation Outline & Proposal Finalization	20 Jan 2026 – 07 Feb 2026	Literature survey on requirement traceability, semantic code understanding, and agentic AI. Finalization of dissertation outline and problem formulation. Submission of outline.
Requirement Analysis & System Architecture Design	08 Feb 2026 – 28 Feb 2026	Design of the cloud-hosted agentic framework. Identification of integration requirements with ITSM, Code Repositories, and DevOps artifacts. Definition of semantic reconstruction pipeline.
Prototype Development Phase I (Artifact	01 Mar 2026 – 20 Mar 2026	Implementation of tools for issue tracking and version control systems. Extraction of commits, tickets, pull request context, and traceability of

Integration Layer)		metadata.
Midterm Report Preparation & Submission	21 Mar 2026 – 28 Mar 2026	Documentation of progress, architecture, preliminary results, and prototype status. Submission of midterm dissertation report.
Prototype Development Phase II (Agentic AI Reasoning Module)	29 Mar 2026 – 20 Apr 2026	Development of multi-agent semantic reasoning for intent reconstruction. Implementation of conversational explanations and requirement-to-code mapping generation.
Testing, Validation & User Evaluation	21 Apr 2026 – 05 May 2026	Functional testing of the system on production-grade repositories. Evaluation of onboarding acceleration, cognitive load reduction, and traceability accuracy.
Final Dissertation Writing & Review	06 May 2026 – 11 May 2026	Consolidation of findings, final documentation, proofreading, and incorporation of supervisor feedback. Completion of final dissertation manuscript.
Final Submission	12 May 2026	Final review and official submission of the dissertation report.

6. Literature References

The state of the art forms the foundation for any successful research project. In the current dissertation, the literature is inclined towards the emerging domain of **requirement traceability**, **semantic program comprehension**, and **AI-assisted software engineering**, particularly focusing on the reconstruction of implementation intent from enterprise development artifacts such as issue tracking systems, version control histories, and collaborative DevOps workflows. The following journals, conference proceedings, and scholarly works are referred from the preliminary literature review.

[1] J. L. C. Guo, J.-P. Steghöfer, A. Vogelsang, and J. Cleland-Huang. *Natural language processing for requirements traceability*. arXiv Preprint, *arXiv:2405.10845*, 2024.

[2] B. Wang, Z. Zou, H. Wan, et al. *An empirical study on the state-of-the-art methods for requirement-to-code traceability link recovery*. Journal of Systems and Software, 2024.

[3] S. J. Ali, V. Naganathan, and D. Bork. *Establishing traceability between natural language requirements and software artifacts using retrieval-augmented generation and large language models*. In Proceedings of the Model Engineering Conference (ER'24), 2024.

[4] S. Ahmad, N. Agha, B. Ray, et al. **Transformers for Code: A Survey**. ACM Computing Surveys (CSUR), Vol. 55, No. 10s, Article 210, 2023

[5] P. Kanade, Y. Agrawal, S. de Melo, et al. **Self-Training Improves Code Generation**. ICLR 2023.

7. Particulars of the Supervisor and Examiner

	Supervisor	Additional Examiner
Name	Hanumanthu Indrakanti	Deepan Kumar
Qualification	M.Tech in Computer Science and Systems Engineering	M.Tech in Data Science and Engineering
Designation	Director of Engineering	Staff Engineer
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8. Remarks of the Supervisor

The proposed study and dissertation project appear to be highly promising and relevant to the current research advancements in intelligent software engineering and requirement traceability. The student has demonstrated a strong understanding of the foundational literature in areas such as cloud computing, cloud computing tools, semantic program comprehension, DevOps artifact analysis, and agentic AI-based developer assistance. The selected topic reflects an important and emerging domain of research, addressing a critical challenge faced in modern enterprise-scale and open-source software development environments.

The dissertation is well justified at the master's level, as it integrates state-of-the-art concepts including cloud-hosted AI architectures, multi-agent reasoning frameworks, and requirement-to-code traceability reconstruction in production-grade codebases. The expected outcome of this work is both technically significant and practically impactful, particularly in accelerating developer onboarding, improving maintainability, implementations, enhancements of product and enhancing contextual understanding of implementation intent.

Furthermore, the inclusion of prototype development and evaluation strengthens the validity of the research methodology, ensuring that the system is assessed under realistic software engineering conditions. This has strong potential for future industrial publication.

After the initial discussions with the student, I have formed the impression that the student is diligent, motivated, and possesses the technical capability required to implement the proposed framework in accordance with standard research paradigms. Therefore, I approve this dissertation project and support its progression under my and additional examiner supervision.

Information about the Supervisor:

The dissertation work is conducted under the supervision of **Mr. Hanumanthu Indrakanti**, M.Tech in Computer Science and Systems Engineering, who currently serves as the **Director of Engineering** at **Aqueralabs India Pvt Ltd, Bangalore**. His expertise in enterprise software systems and cloud-native engineering provides valuable guidance for this research. The work is also supported by an additional examiner, **Mr. Deepan Kumar**, M.Tech in Data Science and Engineering, working as a **Staff Engineer** at the same organization. Both contribute significantly to ensuring the academic rigor and industrial relevance of the dissertation.

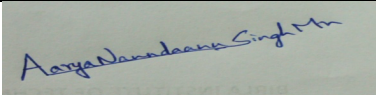
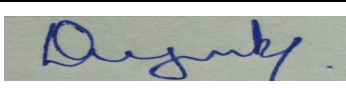
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WORK INTEGRATED LEARNING PROGRAMMES (WILP) DIVISION
SECOND SEMESTER OF ACADEMIC YEAR 2025-2026

CCZG628T : DISSERTATION OUTLINE

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DISSERTATION TITLE	Semantic Reconstruction of Software Intent: A Cloud-Hosted Agentic AI Solution for Study of Requirement Traceability in Production Grade Codebases

Please prepare the outline as a separate document with the following sections along with the above identification information.

1. Cover Page with Student ID No., Name, Course Number, Course Title and Dissertation / Project / Project Work Title, Broad Academic Area of Work.
2. Background (Relevance of the Project to the current work environment in the employing organization)
3. Objectives
4. Scope of Work (To be done by the student independently)
5. Plan of Work (Work to be done during the semester)
6. Literature References
7. Particulars of the Supervisor and Additional Examiner
8. Remarks of the Supervisor

		
Signature of Student	Signature of Supervisor	Signature of Additional Examiner
Name: Aarya Nanndaann Singh M N	Name: Hanumanthu Indrakanti	Name: Deepan Kumar